

Spatial Agents: A Timbre-Driven Ambisonic Instrument for Live Electronics in Csound and Max

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Abstract. *Spatial Agents* is a live-electronics instrument built on a single premise: that the place of a live sound should follow its timbre. Five piezoelectric contact microphones on a hand-built resonating box are analysed in real time into five audio descriptors, and those descriptors move five autonomous sources, or agents, through an Ambisonic field decoded to a horizontal loudspeaker ring or to headphones. The paper is as much about a compositional stance as about a system. Following a tradition of spectral and timbre spatialisation, we treat space not as a container for sound but as a dimension the performer shapes directly: a still, pure resonance settles near the centre of the field, a bright gesture rises, a noisy evolving texture is thrown to the periphery and set spinning. We describe how the descriptors map to motion, focusing on a mapping in which spectral stillness and purity pull a source inward while spectral change pushes it outward, and we explain why we run Csound as the engine inside Max rather than Csound alone: the pairing joins Csound’s analysis and synthesis vocabulary and its portable, sample-accurate processing to the live routing, control and visual feedback that a performance patch needs. The Ambisonic decoder uses the ACN/SN3D convention and is rebuilt at run time from a description of the loudspeaker layout. The approach adapts a descriptor-driven spatialization model previously introduced for corpus-based improvisation; what is new here is a self-contained Csound engine, built from scratch and driven live by a physical contact-microphone instrument rather than a pre-analysed corpus. The instrument is functional and has been validated; we close with the compositional questions it opens and with work still in progress.

Keywords: Ambisonics, timbre spatialisation, live electronics, audio descriptors, Csound, Max

1 Space as a Medium, Timbre as its Handle

Spatial music has long treated the position and motion of sound as a compositional dimension in its own right, and not merely as a way of placing an otherwise fixed object [3]. One strand of this practice binds space to spectrum: instead of moving a whole sound as a point, it distributes or steers the sound according to its spectral content, so that timbre and place become facets of one gesture. Normandeau names this *timbre spatialisation* and makes the medium itself the subject of the work [1]; related frequency-domain and spectral-splitting techniques share the same intuition. More recent work frames timbral spatialisation as a real-time, algorithmic process that derives the motion of spectral material directly from audio analysis [2], the strand to which *Spatial Agents* belongs. Where much of that work distributes the spectral bands of a single sound across the array, or spatialises a pre-analysed corpus, *Spatial Agents* instead moves whole sources as autonomous agents in a live, instrumental setting: a performer plays a resonating box, and the timbre of what they play is what moves each source in space.

The instrument reads five contact microphones on the box, reduces each to a small set of perceptual descriptors, and uses them to drive five autonomous sources through a second-order Ambisonic field. The mapping is chosen so that the spatial image mirrors the sound. A sustained, pure resonance is heard as a still point near the listener; a bright attack rises; a noisy, quickly changing gesture is scattered outward and set in motion [14]. Because the ear already tends to fuse spectral character with apparent location, coupling the two by design gives the performer a spatial handle that feels continuous with the act of playing, rather than a separate layer of control operated from a mixing desk.

This fusion is not incidental. Auditory scene analysis holds that a coherent spectrum is heard as one source in one place [4], which is why steering a sound by its spectral features reads as natural rather than arbitrary. Timbre spatialisation has used this mostly in fixed media, decomposing a recorded spectrum across an array. *Spatial Agents* moves the same principle into live performance: the descriptors follow the performer’s gesture as it happens, so the spatial result is not applied to a recording afterwards but produced in the moment, as a consequence of playing.

Understood this way, the spatial form is not written out in advance but emerges from a rule-based reading of the live sound. In this sense *Spatial Agents* is a small real-time compositional machine, a rule-based system in which perceptual properties of sound generate spatial form; it joins Koenig’s concern with constructed sonic processes to Tenney’s attention to audible perceptual continuity, the figures this edition of the conference invites us to engage. The descriptor-to-space mappings used here were first developed for corpus-based improvisation, where a circular spatialization module drove co-creative agents from spectral descriptors [13]. *Spatial Agents* rethinks and adapts that approach for a live instrument: its contribution is the reimplementation, from scratch, as a single self-contained Csound engine hosted in Max and played from a physical contact-microphone controller, together with an account of why that pairing, rather than Csound alone, suits live performance.

2 From Timbre to Space

The signal path is shown in Fig. 1. Each source is analysed at control rate into the five descriptors of Table 1. Energy is a smoothed **rms** reading; a transient is detected by comparing a fast and a slow **follow2** envelope; the spectral centroid is taken with **pvscent**; spectral spread is the standard deviation of the magnitude spectrum about the centroid, computed with **pvsftw**; and spectral flux is the summed frame-to-frame magnitude difference. These are standard perceptual descriptors of timbre [11].

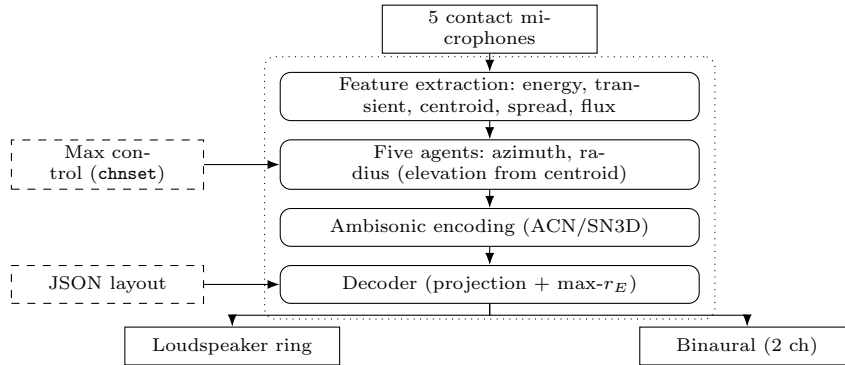


Fig. 1. Signal flow. Five contact microphones are analysed and spatialized inside one Csound engine (the dashed box) hosted in Max; the B-format bus is decoded to a loudspeaker ring or to headphones.

Two mappings hold across the instrument, and stand for its guiding idea. The centroid always controls elevation, so a brighter sound rises; this height is reproduced in the binaural render and on periphonic layouts, whereas a horizontal ring necessarily flattens it into the plane. Spread always controls apparent width by lowering the gains of the higher-order Ambisonic components, so that a pure tone becomes a compact point and a noisy one a diffuse image. The other descriptors act through nine selectable trajectory modes (orbit, pendulum, spiral, Lissajous, random walk, target, jump on transient, a flocking mode in which the agents attract and repel one another, and the mapping below).

The conceptual centre of the instrument is the adaptation of that mapping to live instrumental control [13], in which the spatial behaviour is made to mirror the spectral behaviour of the source. Writing $\hat{f}$ and $\hat{s}$ for the normalized flux and spread, we define a centrality $c = (1 - \hat{f})(1 - \hat{s})$ and set the source radius to $r = 1 - c$, while the azimuthal drift grows with flux. A still, pure sound has $c \approx 1$, so it sits at the centre and barely moves; a changing, noisy sound has $c \approx 0$ and is thrown to the periphery, where it circulates. Since the radius also attenuates and darkens a distant source, a sustained sine tone is heard as a still, present point and an evolving noisy texture as a receding,

Table 1. The five descriptors and the spatial parameter each one drives.

Descriptor	Opcode(s)	Spatial mapping
Energy (RMS)	rms	Rotation speed
Transient density	follow2 (fast/slow)	Direction reversal, jumps
Spectral centroid	pvscent	Elevation (always active)
Spectral spread	pvsftw	Source width
Spectral flux	pvsftw + tables	Drift and centrality

restless orbit. The performer does not aim at coordinates; they play a timbre, and the space follows (Fig. 2).

In Csound this is a handful of control-rate lines:

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kStill    = 1 - kFluxN      ; spectral stillness
kPurity   = 1 - kSpreadN    ; spectral purity
kCentral  = kStill * kPurity ; still + pure -> centre
kRadius   = 1 - kCentral    ; complex/noisy -> periphery
kDrift    = kFluxN * 2      ; more change -> faster drift

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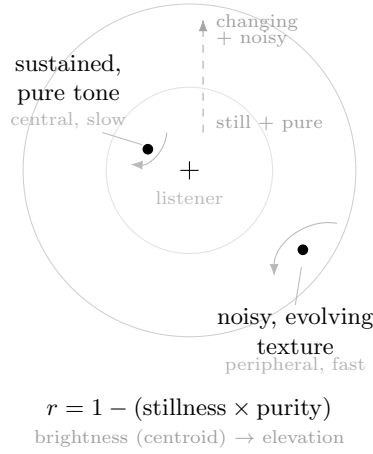


Fig. 2. The sonification mapping. A still, pure sound stays near the centre and drifts slowly; a noisy, evolving sound is pushed to the periphery and orbits quickly, while brightness raises it in elevation.

3 Realisation: Csound inside Max

We run Csound as the engine inside Max [12] through the `csound6~` object [10], and the choice is deliberate. The two environments answer different needs. Csound supplies the analysis and synthesis vocabulary the instrument depends on, from the descriptor opcodes above to user-defined Ambisonic opcodes [7], together with sample-accurate control-rate processing and an engine that is a single, portable text file [9]. Max supplies what a live performance needs around that engine: multichannel hardware input and output, a control surface, on-stage visual feedback of where the agents are, run-time loading of the loudspeaker layout, and integration with the rest of a performance patch. Csound alone would give the processing but not the live ergonomics; a Max patch alone would give the ergonomics but not Csound’s vocabulary and precision. The instrument lives in the pairing, and the paired form is what makes it playable on stage.

Encoding and decoding are user-defined opcodes evaluating the real ACN/SN3D spherical harmonics up to third order, second order being used in the horizontal-ring performances reported here. For these regular rings the decoder is a projection solution with $\max-r_E$ weighting, loudness-normalised by sweeping a virtual source over the horizontal circle. Keeping the ACN/SN3D convention throughout [5] lets us hand a plain B-format stream to an external all-round decoder [6] when the internal

one is not adequate, for instance on an irregular dome, as other recent Csound work also recommends [8]. The decoder is rebuilt at run time: a reader in Max parses a description of the layout and sends it to Csound, and an internal score event recomputes the decoding matrix without interrupting the audio. For headphone monitoring we use a file-free binaural model, avoiding any dependence on loaded impulse responses.

A few practical points make this combination reliable and may save others time. Audio is read and written in blocks rather than per channel; control-rate values leave Csound with a tag and are unpacked with a single route object; the Csound control period must divide the host vector size; and the engine begins only on an explicit start message. The microphone-to-channel wiring was settled empirically, since the object's first inlet carries both control messages and the first audio channel. None of this is exotic, but none of it is documented, and each point cost us time before the instrument played cleanly.

4 Conclusion and Perspectives

Spatial Agents is functional: the five contact-microphone inputs, internal test sources and the full spatial chain have been validated on an eight-loudspeaker ring and on headphones, with five simultaneous spectral-analysis streams running reliably in real time on the M2. The Max layer lets trajectory modes be switched and loudspeaker layouts reloaded live, the decoder being rebuilt without interrupting the audio. More than a spatialisation utility, it is an attempt to give a familiar compositional dimension back to the hands of a live performer. In that sense the medium really is the space [1]: what the instrument affords is a way of hearing spectral character as spatial energy, decided moment by moment by how the box is played.

In its current form the instrument is played live. As a closing, compositional stage, rather than during the live act, we are letting the agents' trajectories be written to files, so that a spatial gesture captured in performance can be reopened and reworked afterwards. Further work concerns recallable parameter scenes, calibration of the descriptors against the physical contact microphones, and the dramaturgy of the performance itself. We hope the account of pairing Csound with Max is useful in its own right, as one practical route to using Csound as a real-time spatialisation engine on stage.

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